



CO₂ Emission from Vehicles in Canada

**Analysis of Vehicle CO₂ Emissions Using Multiple Linear and
Logistic Regression**

Project Report

Group 07

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1. Introduction

Carbon dioxide (CO₂) is a natural and an important part of earth's atmosphere. It is a gas that helps plants to breathe through photosynthesis and it acts as a blanket to keep the planet warm. Even though it makes up only about 0.04% of the atmosphere, it is very crucial for the balance in the environment. For decades human activities, especially the burning of fossil fuels have increased CO₂ concentration to a level that can be very dangerous to the environment. Also, with the growing population the number of vehicles on the road has increased rapidly. Most of these vehicles are powered by gasoline or diesel. This trend is clearly noticeable in developing countries. As some of the results of rising CO₂ levels we are seeing global warming, melting glaciers, rising sea levels, and more frequent natural disasters. In response governments and industries are pushing towards sustainable solutions. The automotive sector is investing in cleaner technologies and promoting zero-emission vehicles. However, the shift towards cleaner transport is slow and still faces challenges such as high costs, limited infrastructure, and consumer acceptance. For our project our goal was to understand the underlying factors that drive emissions from vehicles using real world data from the Canadian government. We are using regression analysis to figure out which internal attributes of a vehicle are associated with vehicle CO₂ emission. By analyzing these patterns, we hope to provide insights and suggest more effective ways to reduce our carbon footprint.

About the data set,

For our project we used a dataset from the Canadian government's fuel consumption ratings program. It provides assessments of fuel consumption and measured CO₂ emissions for fuel-powered commercial vehicles in Canada. The dataset details how CO₂ emissions vary across different vehicle features. It spans a period of 7 years, from 2014 to 2020, and contains a total of 7385 records across 12 columns. Data is available for 2,053 different car models from 42 car brands. The dataset includes attributes such as vehicle make, model, engine size, fuel type, transmission, fuel consumption in city and highway driving, and the corresponding CO₂ emissions.

This dataset is available on both Canadian Open Government Data portal and Kaggle:

<https://www.kaggle.com/debajyotipodder/co2-emission-by-vehicles> (Used dataset)

<https://open.canada.ca/data/en/dataset/98f1a129-f628-4ce4-b24d-6f16bf24dd64#wb-auto-6> (Original dataset)

Objectives of the study,

The main purpose of this study is to identify the factors that are associated with CO₂ emissions from vehicles by examining internal attributes such as engine size, number of cylinders, fuel type, transmission, and fuel consumption in city and highway driving.

Other objectives include:

- Performing descriptive analysis to determine which internal factors have the most impact on vehicle CO₂ emissions.
- Building a multiple linear regression model to predict the amount of CO₂ emissions of a vehicle based on its features.
- Developing a binary logistic regression model by classifying CO₂ emissions into two categories: High and Low.

2. Literature Review

The Canadian vehicle CO₂ emissions dataset has been the focus of many recent studies aimed at developing accurate prediction models. Most of these works concentrate on regression-based approaches that handle both categorical and continuous independent variables. Gurcan (2024), in PeerJ Computer Science, conducted one of the most detailed regression analyses using this dataset. The study applied 18 algorithms across traditional machine learning, ensemble learning, and deep learning regression models. Model performance was evaluated using R², Adjusted R², root mean square error (RMSE), and runtime. Among all tested methods, ensemble models achieved the best results. They consistently reached R² values close to 99.00 with very low RMSE, showing very high predictive accuracy. These models were able to capture non-linear relationships between engine size, different types of fuel consumption (combined, city, highway), cylinder count, and CO₂ emissions. Deep learning models also performed well but they required more computational power and longer training times. According to Gurcan's research most important predictors were fuel consumption variable and engine size, making them crucial for emissions forecasting and sustainability-focused policy design. In a study published in Sustainability in 2023, Abdullah H. and Theyazn H. explored deep learning architectures designed for regression tasks. They used Long Short-Term Memory (LSTM) and Bidirectional LSTM (BiLSTM) models on the same dataset. Model performance was assessed using mean square error (MSE), RMSE, Pearson's correlation coefficient, and R². The BiLSTM model achieved a R² of 93.78, performing better than the standard LSTM. Strong correlations were again found between CO₂ emissions and factors such as fuel consumption and engine size. Gazi Mohammad (2025) in Scientific Reports developed a lightweight multilayer perceptron regression model called CarbonMLP. This model achieved an R² of 99.38 and an MSE of 0.0002, placing it among the most accurate models applied to this dataset. A key feature of this research was the use of eXplainable Artificial Intelligence (XAI) through SHapley Additive exPlanations (SHAP). SHAP made it possible to understand the contribution of each feature to the predictions. Engine size, city and highway fuel consumption, and vehicle type were identified as the most influential factors. Several Jupyter notebooks on Kaggle and GitHub have also applied regression techniques to this dataset. Across these notebooks, a consistent methodology is followed. This includes detailed Exploratory Data Analysis (EDA), data preprocessing to handle outliers and missing values and the comparison of different machine learning models. The analyses explored a range of regression algorithms, from traditional Simple and Multiple Linear Regression to more advanced techniques such as Ridge and Lasso regularization as well as ensemble methods like Random Forest and deep learning models. Engine size, fuel consumption, and cylinder count were the features that consistently identified as the most influential factors in predicting CO₂ emissions. Across all approaches, engine size, fuel consumption, and cylinder count consistently appear as the most important predictors. These findings match the project's goal of building interpretable regression models and provide a solid basis for selecting methods, choosing features, and balancing accuracy with efficiency in predicting CO₂ emissions for sustainable transportation strategies.

3. Descriptive Analysis

Variable definitions

Variable name	Variable description	Data Type
Make	Manufacture or Brand of the vehicle	Categorical
Model	specific design and name given to a vehicle by its manufacturer. 4WD/4X4 = Four-wheel drive, AWD = All-wheel drive FFV = Flexible-fuel vehicle, SWB = Short wheelbase LWB = Long wheelbase, EWB = Extended wheelbase	Categorical

Vehicle Class	Class of vehicle depending on their utility, capacity and weight There are 16 types of different vehicle classes in the dataset such as Compact, Mid-Size, Large...	Categorical
Engine Size(L)	Size of engine used in Liter(L)	Continuous
Cylinders	Number of cylinders (number of combustion chambers in the engine)	Discrete
Transmission	Transmission type with number of gears A = Automatic, AM = Automated manual, AS = Automatic with select shift, AV = Continuously variable M = Manual, 3 - 10 = Number of gears	Categorical
Fuel Type	Type of Fuel used. X = Regular gasoline, Z = Premium gasoline, D = Diesel, E = Ethanol (E85) , N = Natural gas	Categorical
Fuel Consumption City (L/100km)	amount of fuel a vehicle uses to travel 100 kilometers in city driving conditions.	Continuous
Fuel Consumption Hwy (L/100km)	amount of fuel a vehicle uses to travel 100 kilometers in Highway driving conditions.	Continuous
Fuel Consumption Comb (L/100km)	Combined fuel consumption rating in L/100 km. The combined rating reflects 55% city and 45% highway driving.	Continuous
Fuel Consumption Comb mpg	Combined fuel consumption rating in miles per gallon. The combined rating reflects 55% city and 45% highway driving.	Continuous
CO2Emissions(g/km)	The tailpipe emissions of carbon dioxide (in grams per kilometer) for combined city and highway driving	Continuous

Variable Transformation and Categorization

Some categorical variables in the dataset had many detailed categories. While these specific category levels provide very specific detailed information, including all of them in the model would have created many dummy variables. This would make the model complicated and difficult to explain. To address this, the detailed categories were grouped into broader, meaningful groups as follows,

Vehicle class: Originally, the dataset contained 16 different categories. These were grouped into 3 categories (Small, Medium, Large) based on the vehicle's physical size and number of seats.

New category	Description	Includes categories
Small	These are smaller, typically more fuel-efficient vehicle	COMPACT, SUBCOMPACT, MINICOMPACT, TWO-SEATER
Medium	These represent larger passenger cars and station wagons with moderate size and fuel efficiency, distinct from both small cars and larger vehicles	MID-SIZE, FULL-SIZE, STATION WAGON - SMALL, STATION WAGON - MID-SIZE

Large	These are larger, generally less fuel-efficient vehicles	SUV - SMALL, SUV - STANDARD, PICKUP TRUCK - SMALL, PICKUP TRUCK - STANDARD
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Fuel Type: Originally, there were 5 different categories (X, Z, D, E, N). These were combined into 3 categories (Gasoline, Diesel, Natural Gas) based on similar characteristics in energy source. Natural gas is removed because it has only one observation.

New category	Description	Includes categories
Gasoline	The gasoline fuels include Regular (87-88 octane), and Premium (91-94 octane).	X = Regular gasoline Z = Premium gasoline
Diesel	Diesel fuel type	D = Diesel
Ethanol	Natural gas	E = Ethanol (E85)

Transmission Type:

New category	Description	Includes categories
Automatic	A type of gearbox in a vehicle that automatically changes gear ratios, allowing the driver to focus on steering and braking without needing to manually shift gears.	A4, A5, A6, A7, A8, A9, A10, AS4, AS5, AS6, AS7, AS8, AS9, AS10
Manual	A type of vehicle transmission where the driver manually selects and changes gears using a gear lever (shifter) and a clutch pedal.	M5, M6, M7
Variable	A type of automatic transmission that offers a smooth, stepless range of gear ratios instead of fixed gears.	AV, AV6, AV7, AV8, AV10
Automated Manual	An Automated Manual Transmission (AMT) combines the simplicity of a manual gearbox with the convenience of automatic gear shifting.	AM5, AM6, AM7, AM8, AM9

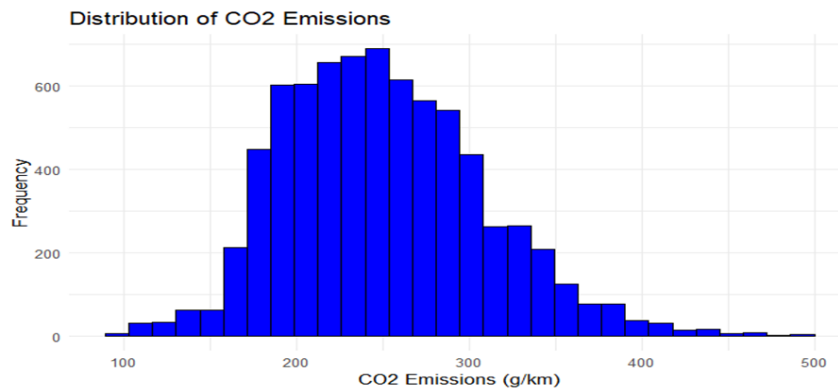
Variable selection and reduction for the analysis

The make of a car (vehicle make) refers to the company or brand that manufactures the vehicle like Toyota, Ford. The car's make can influence the design and styling of the vehicle. The model of a car (vehicle model) refers to a specific version or variation of a vehicle within a particular manufacturing company and describes the features and characteristics such as engine size and other specifications. In this data set, characteristics like number of cylinders in the engine, engine size, fuel type, transmission type, vehicle class and fuel consumption rate are already provided in detail. Therefore, vehicle make and vehicle model were not included in the regression model, as they describe the same underlying characteristics, already represented by variables such as engine size, number of cylinders, transmission type and vehicle class. Removing them helps simplify the model without losing important information.

In addition to that, Fuel Consumption City (L/100km) and Fuel Consumption Hwy (L/100km) were not used to build the model because CO2Emissions(g/km) is measured using Fuel Consumption Comb (L/100km). Fuel Consumption City (L/100km) and Fuel Consumption Hwy (L/100km) are used only to calculate Fuel Consumption Comb (L/100km). fuel Consumption Comb mpg was also not used in the model because it is another representation of Fuel Consumption Comb (L/100km).

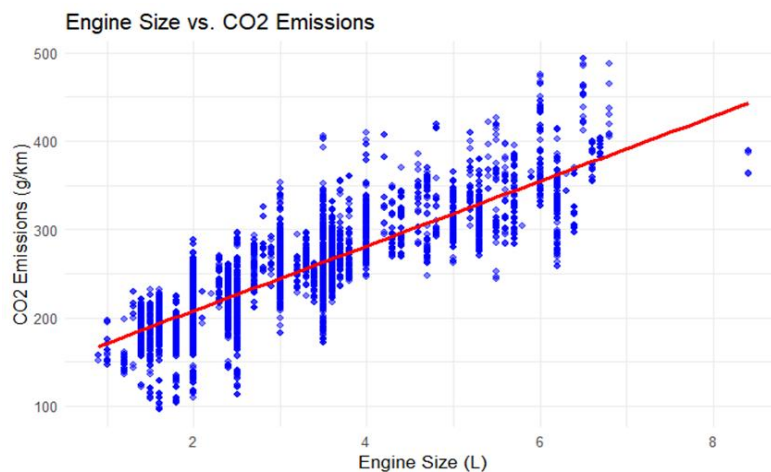
For the analysis, the variables engine size(L), fuel type, transmission type, vehicle class, number of cylinders, Fuel Consumption Comb (L/100km) and CO2Emissions(g/km) were considered.

Exploratory Data Analysis



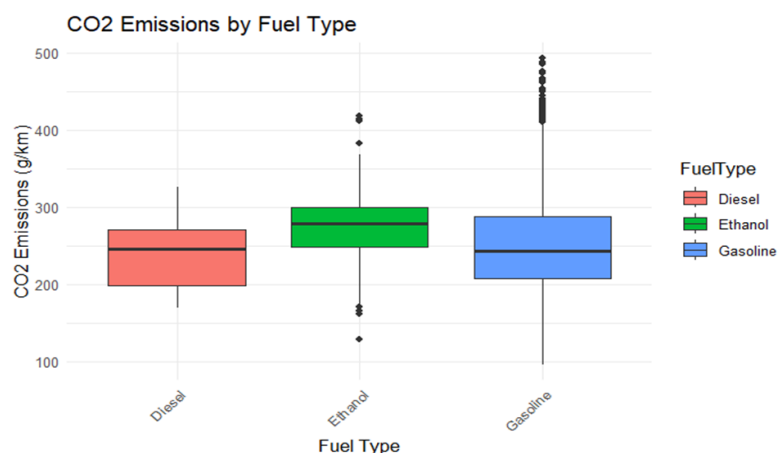
The histogram shows the frequency of vehicles at different levels of CO2 emissions. The distribution is skewed to the right, with a long tail extending towards higher emission values. The most frequent emission range is approximately between 225 and 250 g/km. The majority of vehicles fall within the 200 to 300 g/km range. As emissions

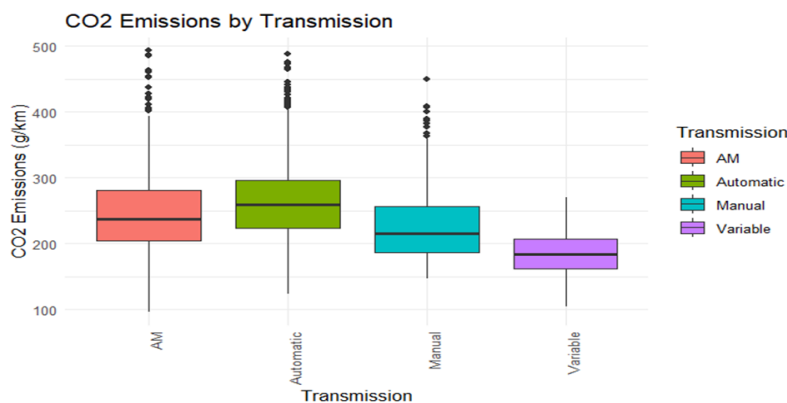
increase beyond 300 g/km, the frequency of vehicles decreases significantly. A very small number of vehicles emit over 450 g/km.



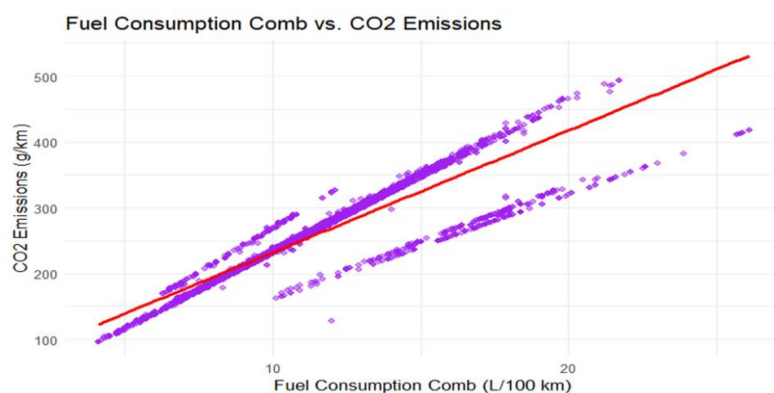
There is a strong positive linear relationship between engine size and CO2 emissions, as shown in the scatter plot. As engine size increases, CO2 emissions also tend to increase. The red trend line shows this positive correlation.

Based on the boxplot, Ethanol powered vehicles have the highest median CO2 emission, and it is approximately 275 g/km. Vehicles using diesel and gasoline have lower median around 250 g/km. Even though the median CO2 emission for diesel and gasoline are alike, the gasoline category shows a wider spread of emissions. This indicates that gasoline powered vehicles have a wider range of CO2 emissions.



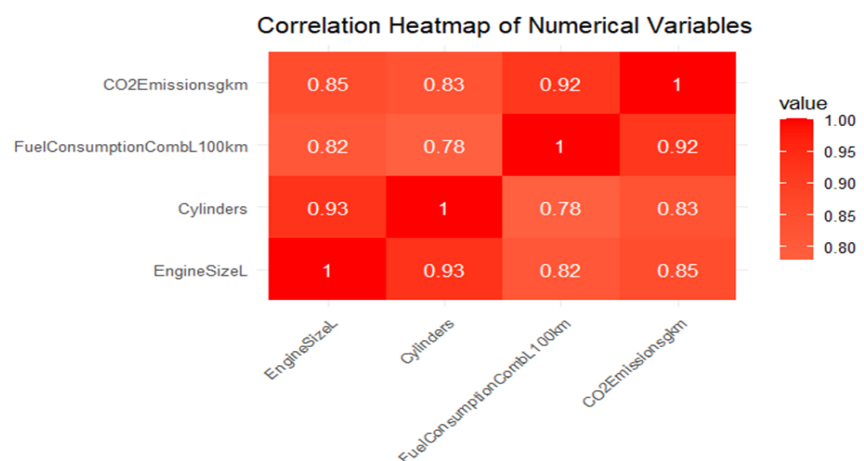


This boxplot shows CO2 emissions by transmission type. Vehicles with automatic transmissions have the highest median CO2 emissions. Manual and AM (Automated Manual) transmissions have similar median emissions that are lower than automatic transmissions, while vehicles with Variable transmissions show the lowest median CO2 emissions.



According to the scatter plot, there is a very strong positive linear relationship between combined fuel consumption and CO2 emissions. The data points are clustered along the red line, which indicates that as fuel consumption increases, CO2 emissions also increase. It is known that burning fuel is the main source of CO2 emission of a vehicle. The plot visually confirms that a higher volume of fuel consumption corresponds to a higher volume of CO2 emission.

This correlation heatmap highlights the correlation among the variables Engine Size (L), Cylinders, Fuel Consumption Comb (L/100km) and CO2 Emissions (g/km). Darker shades of red indicate a strong relationship. CO2 emissions have a very strong positive correlation with all other variables. The correlation is highest with Fuel Consumption Comb (0.92), followed by Engine Size (0.85), and Cylinders (0.83). The highest correlation within the entire heatmap is between Cylinders and Engine Size (0.93), it shows an almost perfect relationship between the two factors. The high correlation values indicate that as one variable increases, the others tend to increase as well.



4. Advanced Analysis

Dummy Variables

Level	Dummy Variables	
	$Fuel_D$	$Fuel_E$
Gasoline	0	0
Deisel	1	0
Ethanol	0	1

Level	Dummy Variables	
	$VClass_S$	$VClass_M$
Small	1	0
Medium	0	1
Large	0	0

Level	Dummy Variables		
	$Trans_A$	$Trans_V$	$Trans_{AM}$
Manual	0	0	0
Automatic	1	0	0
Variable	0	1	0
Automated Manual	0	0	1

4.1 Multiple Linear Regression Model

Fitting the Multiple Linear Regression Model

Variable Selection and Model Building

Our focus was on predicting CO₂ emissions (in grams per kilometer) using multiple variables. We considered categorical variables such as Vehicle Class, Transmission, and Fuel Type, alongside quantitative variables including Engine Size (in liters), Cylinders, and Combined Fuel Consumption (liters per 100 km). To model this relationship, a multiple linear regression was fitted.

```
Call:
lm(formula = CO2Emissionsgkm ~ FuelConsumptionCombl100km + as.factor(FuelType) +
    Cylinders + as.factor(Transmission) + as.factor(VehicleClass))
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-62.255  -2.509   0.182   2.088  43.811
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)    8.57518    0.40194   21.334 < 2e-16 ***
FuelConsumptionCombl100km
  22.01761    0.05215   422.166 < 2e-16 ***
as.factor(FuelType)Diesel
  29.20363    0.42482    68.744 < 2e-16 ***
as.factor(FuelType)Ethanol
-111.77227    0.37438  -298.551 < 2e-16 ***
Cylinders       1.03097    0.06780   15.207 < 2e-16 ***
as.factor(Transmission)AM
   0.57029    0.26633    2.141  0.03229 *
as.factor(Transmission)Automatic
   0.54480    0.18999    2.867  0.00415 **
as.factor(Transmission)Variable
  -2.19289    0.29529   -7.426 1.24e-13 ***
as.factor(VehicleClass)Medium
  -0.97910    0.17601   -5.563 2.75e-08 ***
as.factor(VehicleClass)Small
  -1.46768    0.18142   -8.090 6.94e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 5.376 on 7371 degrees of freedom
Multiple R-squared:  0.9915,    Adjusted R-squared:  0.9915
F-statistic: 9.55e+04 on 9 and 7371 DF,  p-value: < 2.2e-16
```

Significant variables were selected using the forward selection method. Starting from a null model, variables were incrementally added based on their statistical significance and the F-value assessed at each step to determine whether the updated model improved significantly. At the conclusion of this selection process, Vehicle Class, Transmission, Fuel Type, Cylinders, and Combined Fuel Consumption emerged as significant predictors, while Engine Size was not.

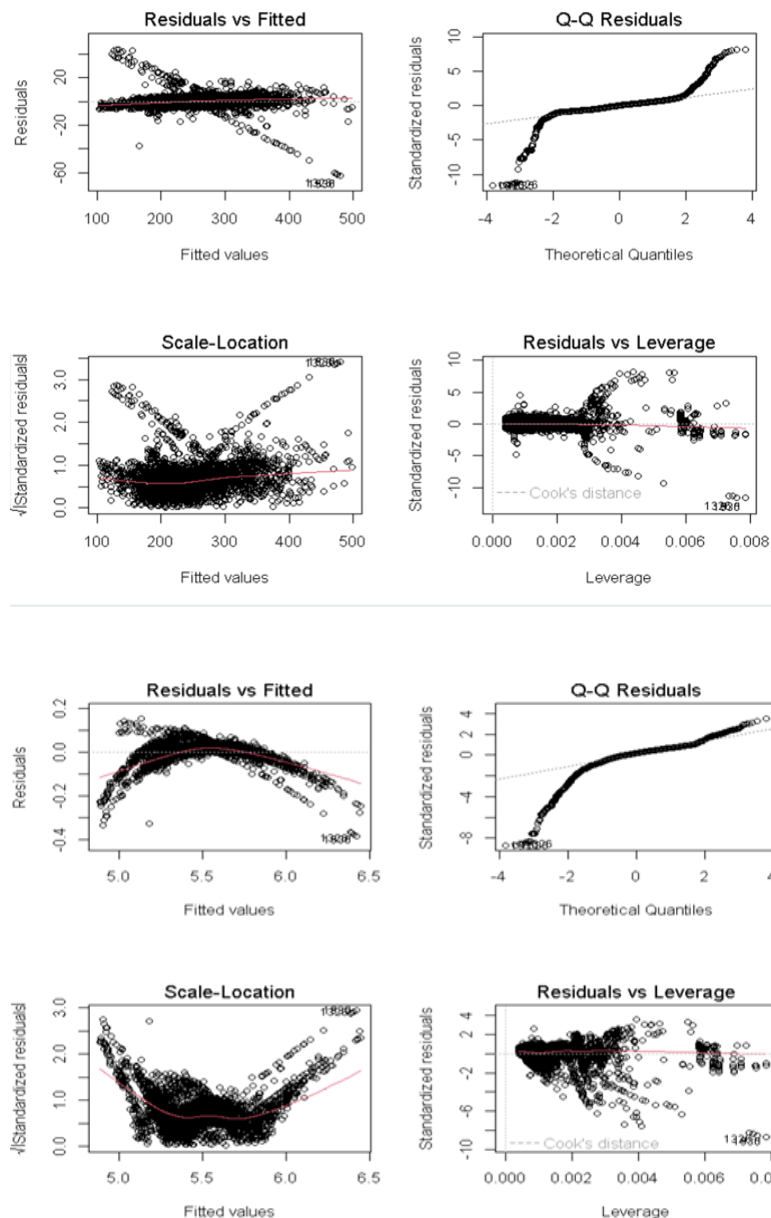
Residual Analysis

- **Residuals vs Fitted plot**

Most residuals are scattered randomly around the zero-line indicating that the model fits the data reasonably well. But there seems to be a very slight upward trend indicating minor deviations from perfect linearity. A few points display a downward trend, but they are limited and do not strongly affect the overall fit. Overall, this plot indicates that the linearity assumption is mostly satisfied though there is room for improvement.

- **Q-Q Residuals plot**

The QQ plot shows that the most of residuals closely follow the reference line indicating an approximately normal distribution but at the extremes, deviations form a S shaped pattern which suggests some departures from normality assumption. We can consider this as sufficient but from this plot as well, it shows that the existing model will benefit from some improvement.



- **Scale-Location plot**

The Scale Location plot shows that most of the residuals are randomly scattered with no strong pattern, but the line shows a very slight U-shaped curvature, but that deviation is minimal. A few points show mild upward and downward trends, but they are also very minimal. Therefore, the assumption of homoscedasticity can be considered to have met.

- **Residuals vs Leverage Plot**

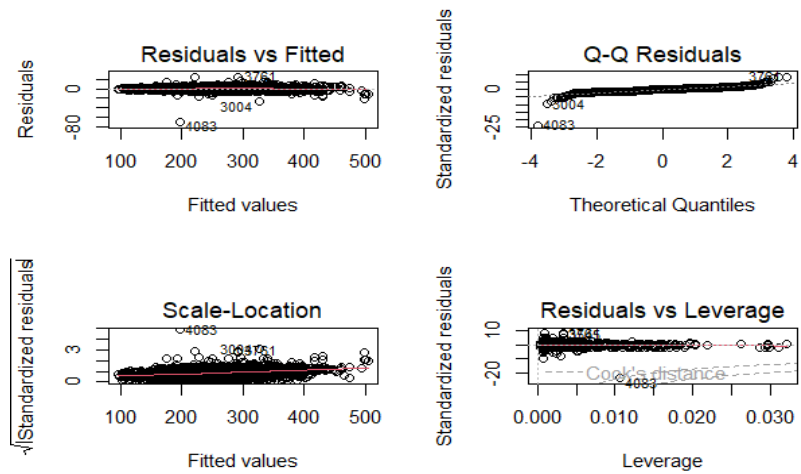
The Residuals vs Leverage plot shows that most observations are randomly scattered around the zero-line showing no strong pattern. But at higher leverage values, the residuals form a shallow U-shaped curve. This suggests that some high leverage points may have an influence on the model.

Because of non-constant variance in the residuals, apply a log transformation to the dependent variable. Even after fitting the model for $\log(\text{CO2Emissionsgkm})$, the model has not improved but instead it has made it worse. Since log

transformation failed, centering variables, adding polynomials were also tried to improve the model but they also failed to improve the model

After the adding interaction terms (FuelConsumptionCombL100km*Fuel Type), residual analysis is

improved. The Residuals vs Fitted plot shows not curved patterns, suggesting non-linearity. The Q-Q plot shows residuals are normally distributed. The Scale-Location plot indicates constant variance indicating approximately flat red line.



```
Call:
lm(formula = CO2Emissionsgkm ~ FuelConsumptionCombL100km + FuelType +
    Transmission + Cylinders + VehicleClass + FuelConsumptionCombL100km *
    FuelType)

Residuals:
    Min       1Q   Median       3Q      Max
-69.263  -2.470   0.642   1.838  24.815

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    1.98802    0.22521   8.827 < 2e-16 ***
FuelConsumptionCombL100km 22.97367    0.02947 779.487 < 2e-16 ***
FuelTypeDiesel  -1.18938    1.29722  -0.917 0.359239
FuelTypeEthanol  3.41447    0.91346  3.738 0.000187 ***
TransmissionAM   0.33176    0.14495  2.289 0.022117 *
TransmissionAutomatic 0.18537    0.10343  1.792 0.073132 .
TransmissionVariable -0.48247    0.16137  -2.990 0.002801 **
Cylinders        0.33498    0.03733  8.973 < 2e-16 ***
VehicleClassMedium -0.48759    0.09604  -5.077 3.93e-07 ***
VehicleClassSmall -0.88219    0.09895  -8.915 < 2e-16 ***
FuelConsumptionCombL100km:FuelTypeDiesel 3.63838    0.14430  25.214 < 2e-16 ***
FuelConsumptionCombL100km:FuelTypeEthanol -7.11238    0.05514 -128.996 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.926 on 7369 degrees of freedom
Multiple R-squared:  0.9975,    Adjusted R-squared:  0.9975
F-statistic: 2.655e+05 on 11 and 7369 DF,  p-value: < 2.2e-16
```

As mentioned previously, after the adding interaction terms the R^2 value has increased compared to the base model. indicating that it explains 99.75% of the variability in the response variable.

Subsequently, the VIF test was done to examine whether there are any significant multicollinearity present among the variables of this final mode. Since all values are below 10 it can be concluded that no significant multicollinearity is present.

	GVIF	Df	GVIF^(1/(2*Df))
FuelConsumptionCombL100km	6.230441	1	2.496085
FuelType	1148.263118	2	5.821171
Transmission	1.563514	3	1.077334
Cylinders	3.966145	1	1.991518
VehicleClass	1.659734	2	1.135036
FuelConsumptionCombL100km:FuelType	1221.653434	2	5.912035

Model Equation

$$(\widehat{CO2Emissions(g/km)}) = 1.98802 + 22.97367 * \text{Fuel Consumption Comb (L/100km)} + 1.18938 * \text{Fuel}_D + 3.41447 * \text{Fuel}_E + 0.33176 * \text{Trans}_{AM} + 0.18537 * \text{Trans}_A - 0.48247 * \text{Trans}_V + 0.33498 * \text{cylinders} - 0.48759 * \text{VClass}_M - 0.88219 * \text{VClass}_S + 3.63838 * (\text{Fuel Consumption Comb (L/100km)} * \text{Fuel}_D) - 7.11238 * (\text{Fuel Consumption Comb (L/100km)} * \text{Fuel}_E)$$

Interpretation of the model parameters

When FuelConsumptionCombL100km increases by one unit, CO₂ emission increases by 22.97, holding other variables constant. Being a Diesel vehicle decreases CO₂ emission by 1.19 compared to Gasoline, while being an Ethanol vehicle increases it by 3.41. Having an AM transmission increases CO₂ emission by 0.33, an Automatic transmission increases it by 0.19, and a Variable transmission decreases it by 0.48, all relative to Manual. Each additional Cylinder increases CO₂ emission by 0.33. Belonging to the Medium vehicle class decreases CO₂ emission by 0.49, while being in the small vehicle class decreases it by 0.88, both compared to large vehicles. Furthermore, the interaction of fuel Consumption with Diesel increases CO₂ emission by 3.64, whereas the interaction of fuel Consumption with Ethanol decreases it by 7.11, holding other variables constant.

4.2 Logistic Regression Model

Fitting the Logistic Regression Model

In this model, **CO2Emissionsgkm** was converted into a binary variable based on its median value.

High (coded as 1)	CO2Emissionsgkm => median (CO2Emissionsgkm)
Low (coded as 0)	CO2Emissionsgkm < median (CO2Emissionsgkm)

Thus, the dependent variable is binary. Logistic regression can be used to predict a binary dependent variable based on continuous and/or categorical independent variables. For fitting logistic regression model, the following assumptions are considered,

- Independent of observations, logistic regression requires the observations to be independent of each other.
- There is not multicollinearity among the predictors.
- logistic regression assumes linearity of independent variables and log odds of the dependent variable.

Variable selection and Model Building

A forward selection method was used to build the model. The process began with the intercept only model (null model) and added predictors step by step. At each stage, variables were tested using reductions in deviance (reduction in AIC value) and statistically significant improvements in the model. The predictor that gave the largest and statistically significant improvement (minimum AIC, minimum deviance of residuals) was included in the model and the process continued until no further variables reduced deviance or AIC. Further at each stage the reduced model was compared with the full model using the difference of the deviance statistics of the full and reduced models, confirming that the selected model was adequate. Finally, the goodness of fit test was used to identify whether the model fits the data well or not.

Then main effect model is the,

```
Call:
glm(formula = response ~ FuelConsumptionCombL100km + as.factor(FuelType) +
    as.factor(VehicleClass) + as.factor(Transmission), family = binomial("logit"))

Coefficients:
(Intercept)                -175.4342    12.0274 -14.586 < 2e-16 ***
FuelConsumptionCombL100km    16.6041     1.1332  14.652 < 2e-16 ***
as.factor(FuelType)Diesel    21.3125     1.6530  12.893 < 2e-16 ***
as.factor(FuelType)Ethanol  -70.7008     4.8498 -14.578 < 2e-16 ***
as.factor(VehicleClass)Medium -1.2783     0.3707  -3.449 0.000564 ***
as.factor(VehicleClass)Small -1.5818     0.3455  -4.578 4.69e-06 ***
as.factor(Transmission)AM     1.4475     0.5364   2.699 0.006960 **
as.factor(Transmission)Automatic 0.3211     0.4364   0.736 0.461919
as.factor(Transmission)Variable -0.5468     0.7623  -0.717 0.473183
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 10230.38  on 7380  degrees of freedom
Residual deviance:  438.42  on 7372  degrees of freedom
AIC: 456.42
```

In this case CO₂Emissionsgkm = High is coded as 1 and CO₂Emissionsgkm = Low is coded 0. In this scenario probability of a vehicle having high CO₂Emissionsgkm is modeled. Here, P represents the proportion for a vehicle having high CO₂Emissionsgkm (CO₂Emissionsgkm = high).

Model Equation

$$\ln\left(\frac{\hat{P}}{1-\hat{P}}\right) = -175.4342 + 16.6041 * \text{FuelConsumptionCombL100km} + 21.3125 * \text{Fuel}_D - 70.7008 * \text{Fuel}_E - 1.2783 * \text{VClass}_M - 1.5818 * \text{VClass}_S + 1.4475 * \text{Trans}_{AM} + 0.3211 * \text{Trans}_A - 0.5468 * \text{Trans}_V$$

Goodness of fit test

H₀ = Model fits the data well H₁ = model does not fit the data well

Test statistic(deviance) = 438.42 degrees of freedom = 7372

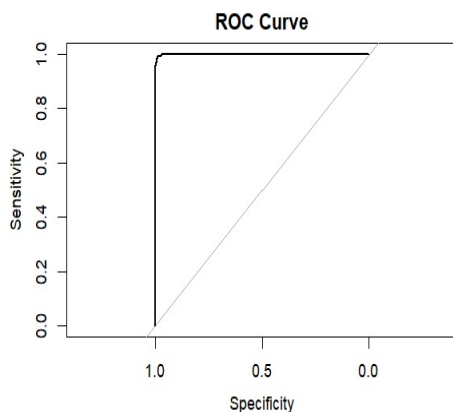
Critical value at 5% significance level = 7572.857

At 5% significance level, test statistic(deviance) < Critical value, do not reject.

There is sufficient evidence to conclude that the model fits the data well at 5% significance level.

Model adequacy checking (model diagnostics methods)

We evaluated the model using the Hosmer-Lemeshow Goodness of Fit Test and ROC Curve (AUC) methods.



For Hosmer-Lemeshow Goodness of Fit Test we got a p-value of 0.9998 is well above the 0.05 threshold indicating that the model fits the data very well.

The ROC curve shows that the model is performing exceptionally well, with very high accuracy and discriminatory ability.

We got an AUC of 0.9994 which is extremely high, approaching 1.0, which indicates near-perfect discrimination. This means the model excellently distinguishes between the two classes of response (high vs. low emissions) across all probability levels.

There is minimum multicollinearity because all GIVF values are less than 10.

```
> vif(logm)
              GVIF Df GVIFA(1/(2*Df))
FuelConsumptionCombL100km 81.395833 1 9.021964
as.factor(FuelType)        83.122732 2 3.019465
as.factor(VehicleClass)    1.884700 2 1.171684
as.factor(Transmission)    1.653974 3 1.087480
```

As there is minimum multicollinearity and the model adequacy checking tests show that the model fits well, no interaction terms were considered.

Interpreting model parameters

- When FuelConsumptionCombL100km by one unit, the log odds of High CO₂ emission increases by 16.6041, holding other variables constant.
- Odds of having high CO₂ emission for those who had used Diesel as the fuel type is times 1.8×10^9 , the odds of having high CO₂ emission for those who had used gasoline, holding other variables constant.
- Odds of having high CO₂ emission for those who had used gasoline as the fuel type is times 5.06×10^{30} , the odds of having high CO₂ emission for those who had used ethanol, holding other variables constant.
- Odds of a vehicle having high CO₂ emission for large vehicle class is 3.5905 times the odds of a vehicle having high CO₂ emission for medium vehicle class, holding other variables constant.
- Odds of a vehicle having high CO₂ emission for large vehicle class is 4.8637 times the odds of a vehicle having high CO₂ emission for small vehicle class, holding other variables constant.
- Odds of a vehicle having high CO₂ emission for transmission AM is 4.2524 times the odds of a vehicle having high CO₂ emission for transmission manual, holding other variables constant.
- Odds of a vehicle having high CO₂ emission for transmission Automatic is 1.3786 times the odds of a vehicle having high CO₂ emission for transmission manual, holding other variables constant.
- Odds of a vehicle having high CO₂ emission for transmission manual is 1.7277 times the odds of a vehicle having high CO₂ emission for transmission Variable, holding other variables constant.
- The intercept estimates of -175.4342 represents the log odds of a vehicle having high co2 emission when all categorical predictor variables are at their reference level (fuel type=Gasoline, vehicle class=Large, transmission type=Manual) and continuous predictor variables (Fuel Consumption Comb (L/100km) =0) are equal to zero.

5. Discussion

Summary

The aim of this project was to identify the factors associated with CO₂ emissions of different vehicle characteristics using multiple regression and logistic regression techniques.

In the multiple regression analysis, Fuel consumption, Cylinders, Fuel Type, Vehicle Class, and Transmission were identified as the most significant predictors. The error analysis showed slight deviations from the model assumptions, therefore methods such as log transformation, centering variables and adding polynomials were attempted but there were no improvements. But including an interaction term improved the model; however, it introduced high VIF, but the interaction term was kept in the final model due to its contribution to the model improvement.

For the logistic regression model, Fuel Consumption, Fuel Type, Vehicle Class, and Transmission were identified as significant predictors. Model adequacy was evaluated using the Hosmer–Lemeshow goodness-of-fit test and the ROC curve, both of which indicated that the model was adequate.

Conclusion

According to the multiple regression model, fuel consumption had the largest positive effect, showing that higher fuel use strongly increases the emission. So, the factors that relate to the fuel consumption such as number of cylinders, some fuel type also tends to increase the CO₂ emission. Diesel fuel shows a negative coefficient suggesting that after keeping other factors constant, diesel may emit less CO₂ than gasoline. This highlights the benefit of developing low emission fuel types. Changing the number of cylinders or Vehicle class is not practical as different vehicles have different purposes making fuel type optimization very important in reducing CO₂ emission.

However, in the logistic regression model, the coefficient for diesel is positive indicating higher odds of being in the high emission category compared to gasoline. This contradiction may be due to how different vehicle classes use different fuel types. The majority of small and medium size vehicles tend to use gasoline thus having smaller emission rates compared to larger vehicles that use diesel fuel that have higher CO₂ emission rates due to other factors such as higher fuel consumption, larger engines and more cylinders. So, the apparent contradiction shows the difference between modeling a continuous outcome vs a categorical outcome.

Overall, the multiple regression results align with the real-world observations as diesel vehicles often emit less CO₂ compared to gasoline. The vehicle's transmission also has an influence on the CO₂ emission as modern complex transmission systems such as Automatic and Automatic Manual tend to have higher CO₂ emission rates compared to Manual when other factors are kept constant (These systems require more energy to function thus resulting in more fuel consumption). Vehicle characteristics such as engine size, cylinders, vehicle class do influence the CO₂ emission but much of their effect is mediated through fuel type and fuel consumption. As an example, a larger vehicle will have a higher fuel consumption and that is due to having a larger engine with more cylinders consuming more fuel.

Challenges and limitations

This study faced several challenges and limitations. The continuous variables, Engine Size, Cylinders and Fuel Consumption are highly correlated leading to multicollinearity. Some categories in the categorical variables also have lower count of records compared to other categories creating class imbalance that could bias model estimates towards the majority group.

More important factors like vehicle weight, aerodynamics, hybrid/electric systems can also heavily influence the CO₂ emission but were not included in the dataset which may have led to an oversimplification of the model.

Error analysis indicated some deviations from the linear regression model assumptions such as minor deviations from residual normality, heteroscedasticity and a few high leverage points which were not resolved through transformations, adding polynomials or centering variables. So, the lack of knowledge

to handle such situations was a great limitation as this caused the model to inhibit the problems leading to inaccurate results even when the model has a high R^2 value.

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Individual Contribution

Name	Index	Contribution
D. A Yahathugoda	s16877	Reduced the number of categories in categorical variables meaningfully. Helped in doing the descriptive analysis. Fitted the multiple linear regression model and did the error analysis and improvements. Helped in fitting the logistic regression model and adequacy checking. In the report: wrote the summary, conclusion and challenges and limitations and helped in improving the advanced analysis section.
Dinusha Priyashan Haputhanthri	s16798	Fitted the multiple linear regression model and logistic regression model with model adequacy testing. Helped to do residual analysis for fitted models. In the report: Wrote the Variable definitions, Variable Transformation and Categorization, Variable selection and reduction, logistic regression model including variable selection and model building ,goodness of fit test and Interpreting model parameters. Helped to write the introduction section.
Ranindu Kariyapperuma	s16810	Conducted the literature review and wrote that section. Fitted the multiple linear regression model and wrote the model building part of it. Also rechecked the logistic linear regression model and model adequacy tests. Create the cover page and the formatting of the document.
Harsha Milinda Dharmasena	s16977	Wrote the Introduction section. Helped to re-check the fitted multiple linear regression model and the logistic linear regression model and model adequacy tests. Contributed to do the descriptive analysis section.
Kavindu Anjana Perera	s16829	Wrote the residual analysis part in advance analysis section. Helped to re-check the fitted multiple linear regression model and the logistic linear regression model with model adequacy tests. Contributed to do the descriptive analysis section.